

Types of addressing mode:

1. Implied mode:

In implied addressing the operand is specified in the instruction itself.

- **Example:** CLC

2. Immediate addressing mode:

In this mode data is present in the address field of the instruction.

- **Example:** MOV AL, 35H

3. Register mode:

In register addressing the operand is placed in one of 8 bit or 16 bit general purpose **register**.

- **Example:** MOV AX, CX

4. Register Indirect Mode

In this addressing mode, the operand's offset is placed in any one of the registers BX, BP, SI, DI, specified in the instruction.

- **Example:** MOV AX, [BX]

5. Direct Addressing / Absolute Addressing Mode

The operand offset is given in the instruction as an 8-bit or 16-bit displacement offset or the operand offset is given in the instruction as an 8-bit or 16-bit displacement element.

- **Example:** ADD AL, [0300H]

6. Indirect Addressing Mode

➤ Register Indirect Mode:

In this mode, effective address is in the register, and corresponding register name will be maintained/mentioned in the address field of an instruction.

- **Example:** MOV A, @R0 → 8051 Architecture

➤ Memory Indirect Mode:

In this mode effective address is in the memory and corresponding memory address will be maintained in the address field of the instruction.

- **Example:** MOV AX, [[3008H]] → Conceptual notation (not valid 8086 syntax)

Index Addressing Modes:

- **Indexed Addressing Mode:** MOV AX, [SI + 05H]
- **Based Indexed Addressing Mode:** ADD AX, [BX + SI]